

SEN381

Milestone 1

Engineering Foundation &
Requirements Baseline

Chanice Grant (601597)

Milenke Field (578122)

Megan Heather Wathen (601611)

Table of Contents

1. Engineering Foundation	2
1.1 Problem and Business Need	2
1.1.1 Current Problem	2
1.1.2 Business Need	3
1.1.3 Intended Business and Stakeholder Value	3
1.2 Stakeholder Analysis	4
1.2.1 Stakeholder Identification	4
1.2.2 Stakeholder Analysis Matrix	5
1.2.3 Stakeholder Conflicts and Competing Expectations	6
1.2.4 Stakeholder-to-Requirement Linkage	6
1.3 Scope Baseline	7
1.3.1 Scope Statement	7
1.3.2 In-Scope	8
1.3.3 Out-of-Scope	9
1.3.4 Deferred and Future Scope	9
1.3.5 Deliberate Scope Exclusions and Deferments	9
1.4 Project Constraints	10
1.4.1 Scope Constraints	10
1.4.2 Schedule Constraints	11
1.4.3 Cost and Resource Constraints	11
1.4.4 Quality Constraints	11
1.4.5 Security Constraints	12
1.4.6 Constraint Interactions and Trade-offs	12
1.5 Engineering Decision Log	13
1.5.1 Purpose of the Engineering Decision Log	13
1.5.2 Milestone 1 Engineering Decisions	13
1.5.3 Deliberately Deferred Decisions	15
2. REQUIREMENTS BASELINE	16
Functional Requirements (FRs):	16
Non-Functional Requirements (NFRs):	18
Acceptance Criteria:	20
Requirements Traceability Matrix (RTM):	0
Traceability Example:	0
3. ENGINEERING CONTROL & FORWARD PLANNING	0

Table 1: Risk Register	0
Introduction:.....	0
Table 2 : CivicConnect Forward Engineering Considerations Register	0
GitHub and Team Governance	1
Table 3: CivicConnect GitHub Governance Controls	1
GitHub Evidence	2
Controlled Change Flow	2
Evidence Screenshots	3
4. PED v1.0 and Baseline Gate/Sign-Off Evidence	6
Document Version History:	6
Baseline Gate & Sign-Off Evidence	7
Functional & Quality Gate Milestones:	7
Formal Sign-Off Matrix:	7
APPENDIX A – AI RESEARCH & VERIFICATION RECORD	8
References	9

1. Engineering Foundation

1.1 Problem and Business Need

1.1.1 Current Problem

The community-based organisation uses various disjointed channels and services to manage requests, such as email, phone calls, WhatsApp messaging, spreadsheets and paper records. The requests may be for facility issues, equipment damage, security, IT services, maintenance, lost items and other operational issues. This disorganized approach has led to duplication of requests, missed requests, misallocated requests and lost requests, and requesters have less awareness whether their requests have been received, assigned, pending, solved or closed.

So, the issue is not just the fact that there are multiple channels for communication, but that there's no single controlled process in which the service request can be managed across its lifecycle. Staff may have difficulty in prioritising requests, working out who is responsible for what and coordinating work, and management might not have clear, consistent data about outstanding, overdue and closed requests. There is also no controlled record of the history of each request, as there are no pre-built fields for reporting that are consistent and manual.

The general concern with the fragmented service processes is supported by research. Fragmented access pathways and fragmented information systems can be part of interconnected organisational barriers that can decrease responsiveness and create challenges for effective coordination, says (Castillo de Mesa, et al., 2026). While not directly focused on CivicConnect, it illustrates that fragmentation can have broader organisational repercussions than just information storage.

The existing method thus presents issues of visibility, coordinating, accountability and traceability for CivicConnect. It's hard to identify who has requested something, what was done, how it progressed and if it was resolved or not if there isn't a controlled lifecycle record.

1.1.2 Business Need

The organisation needs a digital platform for services, which gives a dependable traceable and usable service request submission, management, monitoring and reporting method. CivicConnect should therefore go beyond the digitalisation of the current process. The business requirement is to put in place a well-defined lifecycle for the delivery of the services where the flow of information, responsibility and status is consistent.

It should be more visible to make it easier for the requester and authorised staff to see the status of any request. It should enhance accountability by creating a more transparent chain of command for requests and keeping track of relevant activities and any changes in status. There is a need for more reliable information to track service activity, outstanding/overdue work, and support service-performance analysis in management.

This business need should be approached from a requirement engineering point of view and not directly in a technical design. (Alzayed, 2024) pointed out that the lack of understanding of stakeholder needs, scope creep, and inadequate stakeholder engagement are among the major problems that can impact on the success of e-government projects. The study also revealed that using requirements-engineering practices was positively associated with project success. This underscores the importance of first determining CivicConnect's business problem and their stakeholder needs before making detailed technical decisions.

The organisation needs a solution that will be sustainable in the project's conditions too. Therefore, CivicConnect must not impose an unreasonable burden on the current process in terms of technical, operational or financial burden. The business requirement is, therefore, not for the most features possible, but for a controlled solution, that delivers enough value to the stakeholders, that fits within the project's scope and constraints.

1.1.3 Intended Business and Stakeholder Value

CivicConnect aims to be a valuable tool for several stakeholder groups, who are impacted by the existing service request process.

For the requesters, it's about greater visibility and communication. Requests should be made via a controlled process, the requester should be able to see the status and history of the request and receive meaningful feedback on requests accepted, rejected, updated and completed. This helps to minimize doubt as to receipt and disposition of requests.

Staff are expected to see better co-ordination and control. Authorised staff should be able to find relevant requests, identify and/or accept responsibility for the request; update requests with controlled status transitions; and record actions, comments and resolution information. This will give better ownership and a more standardised method of handling work than what is currently in place.

CivicConnect should offer greater transparency and oversight for stakeholders involved in management and oversight. The capability of identifying open, overdue, resolved and closed requests and analyse information by relevant categories or statuses should facilitate accountability and analysis of service performance.

From an organisational perspective, these stakeholder benefits add up to a more managed and traceable lifecycle of the service request. It's not about replacing paper and communication channels with a digital interface, it's about the end goal: increased visibility, accountability, coordination and access to reliable information.

Finally, CivicConnect is a successful effort if it meets the common needs of its stakeholders as outlined in its scope and constraints. A working app is not enough to ensure that the business problem has been solved. The value of the system, however, must be traceable to stakeholder needs and subsequently requirements, acceptance criteria and evidence of project success.

1.2 Stakeholder Analysis

To identify people or groups that are affected by CivicConnect, understand their needs and expectations, and understand how their expectations impact the needs and scope of the system, stakeholder analysis is used. This is significant, because the various stakeholders engage the service-request lifecycle in various ways, and can have conflicting priorities.

(Alzayed, 2024) states that stakeholder analysis is an important practice in requirements engineering and highlights the use of multiple stakeholders during the entire development lifecycle. The study also points out that lack of awareness of the needs of the stakeholders and lack of engagement with the stakeholders are important causes of e-government project failure. While CivicConnect is not an e-government system itself, these findings are relevant to its requirements-engineering process as they are also relevant to CivicConnect's task of identifying and articulating the needs of various user groups as requirements for a system.

1.2.1 Stakeholder Identification

The stakeholder groups initially identified for CivicConnect are requesters, staff, management/oversight stakeholders, and the organisation that will focus on CivicConnect for the community.

Requests made by the people who put in the service request, and who later need information about the request. Their core needs are having a straightforward and user-friendly way to make a request, knowing that the request was recorded, and knowing the status of the request and what to do when changes occur. As such, their interest in CivicConnect is high since they relate to the system directly and are influenced by the results of the requests they submit.

The staff handles the processing and management of service requests. They want to see requests that are relevant to them, they want to see good search and filtering, they want to see who it's assigned to, they want to be able to record actions, comments, resolution information, and see updates on status. High interest and high influence are achieved as staff must be interested in CivicConnect and have a high level of influence on the platform's effectiveness as part of the operational service-request process.

Information about the activity of the services is more important for management and oversight stakeholders, as they do not deal more with individual requests than requesters do. They need to see open, overdue, closed, and resolved requests, access to information that can be analysed by categories or statuses, and have adequate records for accountability and analysis of service performance. High influence for management: the requirements and controls required for the system can be influenced by management priorities, accountability expectations and operating policies.

The community-based organisation is also a stakeholder as it is responsible for the overall service request process and thus will have the operational, technical and financial implications of the solution. Its overall requirement is to enhance visibility, coordination and accountability of CivicConnect with sustainability under the given resources and project constraints.

1.2.2 Stakeholder Analysis Matrix

<i>Stakeholder</i>	<i>Primary Needs / Expectations</i>	<i>Interest</i>	<i>Influence</i>	<i>Engineering Implication</i>
Requesters	Easy request submission, status visibility, request history and meaningful feedback	High	Medium	Requirements must support a usable submission and tracking process without exposing information the requester is not authorised to access.
Staff	Access relevant requests, search/filter, clear ownership, controlled status changes, action records and resolution functions	High	High	Requirements must support efficient request management while controlling permissions and lifecycle transitions.
Management / Oversight	Reliable service information, identification of overdue/outstanding work, accountability and performance analysis	High	High	The system must preserve sufficient structured and traceable information to support monitoring and reporting.
Community-focused organisation	Controlled service-request process, improved coordination and accountability, sustainable operation	High	High	Scope and later engineering decisions must balance stakeholder value against cost, resources, quality, security and maintainability.

Importance of a stakeholder depends on two factors, as indicated by the matrix, and not on frequency of stakeholder use of the system. In such cases, for instance, management might have fewer of the operational staff's day-to-day request-processing activities, but they have a high influence, since they need information for their accountability and oversight. Likewise, requesters might have lower levels of organizational decision-making power, but higher levels of interest as it directly impacts their experience of the request process.

(Alzayed, 2024) agrees with the stakeholder analysis approach to identify and analyse the needs, concerns and expectations of stakeholders to take different perspectives into consideration in the process of requirements engineering. The research also correlates the comprehensive stakeholder analysis with the understanding of the needs of stakeholders and suggests requirements traceability to ensure the linkage between the needs of stakeholders and requirements.

1.2.3 Stakeholder Conflicts and Competing Expectations

Within CivicConnect, there may be some differences in stakeholder expectations. It is important that these conflicts are identified during Milestone 1 as trying to meet the needs of one stakeholder could impact another stakeholder or impact the project constraints.

There is one important tension between transparency and the control of information in a requester. Requesters can be given detailed information about the progress of their request; however, it is possible for them to see all the information, including internal comments or staff information or sensitive information which should only be visible to authorised users.

CivicConnect must therefore give meaningful feedback to the requester but not assume that anything that is recorded internally is available for the requester to view.

There is another conflict between management and efficiency of staff. Detailed and reliable records allow for better management as they aid accountability and reporting. But having to manually capture information for every action may cause more work for staff and not be as efficient as CivicConnect's vision. The project needs to identify what information is needed for accountability without adding unnecessary overhead to operations.

Also, there may be a conflict between the features that stakeholders want and project constraints. Additional functionality that may be identified by the Requesters, staff or management would be value added, but would result in more work necessary to specify, secure, implement, test, document, deploy and maintain the system. Requests from the stakeholders cannot then automatically be scope that is committed. They are valuable if they fit the schedule, resources, quality and security/risk criteria before they're added.

The following conflicts highlight the need to carry out stakeholder analysis throughout the requirements engineering process and not as an isolated exercise. (Alzayed, 2024) advocates for stakeholder engagement throughout the project life cycle and argues that lack of stakeholder engagement is a factor that leads to incomplete definition of the requirements and project failure.

1.2.4 Stakeholder-to-Requirement Linkage

Stakeholder analysis should not be a standalone project artefact but rather must result in measurable engineering requirements. Thus, each important stakeholder requirement should include a source/justification for one or more CivicConnect requirements.

For example, if the requester needs to know the status of the request, then the requirements should permit an authorised requester to see the status and history of his/her request. Staff should be clear in their ownership leading to requirements for the assignment or acceptance of responsibility for a request. The demand for accountability and direction from management should lead to the provision of requirements for identifying the states of requests and retaining adequate information for the analysis of services' performance.

This establishes a first traceability relationship:

Stakeholder → Stakeholder Need → Requirement → Acceptance Criteria

The detail requirement identifiers and acceptance criteria will be determined in the Requirements Baseline and Requirements Traceability Matrix. The stakeholder analysis is thus not a duplication of those artefacts but supplies them from an upstream perspective.

Keeping this connection is also the basis for subsequent change analysis. When a stakeholder needs changes, the team can decide which stakeholder requirements and acceptance criteria might be impacted rather than altering the system without considering possible impacts downstream. (Alzayed, 2024) refers to requirements traceability as a practice to track the evolution of requirements and make sure they are linked to the needs of stakeholders throughout a project.

1.3 Scope Baseline

The scope baseline serves as the boundaries of CivicConnect for the project and identifies the capabilities that the team will be engineering for the project. It is intended to limit uncontrolled growth of the project and to give a solid foundation to requirements, project planning, risk management and then engineering choices.

Volatility of requirements is especially important when defining a scope baseline. In (Silva, et al., 2024), they explore the concept of requirements volatility in government software projects with multiple stakeholders and demonstrate that the volatility of requirements poses problems that must be actively managed. For CivicConnect, this is helpful for setting an agreed baseline, while still enabling legitimate changes to be considered by means of a controlled impact analysis, not scope expansion.

1.3.1 Scope Statement

CivicConnect will offer a controlled digital platform to manage community service requests from submission to their authorised life cycle. The system will facilitate the requester when preparing and tracking requests, the staff when handling and resolving approved requests and management or oversight stakeholders when they need information to monitor accountability and service performance.

The scope of the commitment is targeted at the following problems: Fragmented handling of requests, limited visibility, unclear ownership and weak traceability. The system will therefore have a central controlled record of service requests and their associated service lifecycle information – not digitising all the operational activities of the organisation.

The scope is deliberately constrained to capabilities which directly add value to the identified business need and stakeholder. Extra functionality will not be added on because it can be or is

desired. Any new baseline scope impact needs to be considered in terms of stakeholder value and its impact on schedule, cost/resources, quality, security and risk before it is accepted.

1.3.2 In-Scope

The following capabilities are included within the initial CivicConnect scope:

<i>Area</i>	<i>In-Scope Capability</i>
Request submission	Allow requesters to submit service requests with the information required for processing.
Request categorisation	Categorise requests using a controlled category mechanism.
Requester tracking	Allow requesters to view the status and history of their submitted requests.
Requester feedback	Provide meaningful feedback when requests are accepted, rejected, updated or completed.
Staff request access	Allow authorised staff to access service requests relevant to their responsibilities.
Search and organisation	Allow staff to search, filter or sort requests using useful criteria.
Responsibility	Support assignment or acceptance of responsibility for requests.
Lifecycle management	Support controlled changes to request status.
Request records	Allow relevant actions, comments and resolution information to be recorded.
Resolution and closure	Allow authorised staff to resolve or close requests.
Management oversight	Allow management to identify open, overdue, resolved and closed requests.
Service information	Provide sufficient structured information for accountability and service-performance analysis.
Access control	Restrict functionality and information according to appropriate user authorisation.
Traceability	Maintain sufficient lifecycle information to provide a controlled and traceable record of a service request.

These capabilities are derived from the minimum business capabilities established by the CivicConnect Master Project Brief rather than from an attempt to maximise the number of system features.

The detailed behaviour of these capabilities will be defined through the functional requirements, non-functional requirements and acceptance criteria rather than within the scope baseline itself.

1.3.3 Out-of-Scope

The following is not within the scope of work for the committed CivicConnect at this time, except as approved in the controlled scope change process:

- Replacement of the organisation's wider communication systems (email, telephone or WhatsApp).
- Control of activities within an organisation other than the service-request lifecycle.
- The physical completion of the maintenance, security, IT support or other requested activity.
- Features that are not needed by a stakeholder or for a project.
- Scope creep is the uncontrolled inclusion of more features and/or functions after the scope baseline, without consideration for their impact downstream.
- Functionality or integrations at production scale that go beyond the requirements and limitations of the academic CivicConnect project but are not warranted.

Managing a request is key to differentiating from the work described in the request. For instance, CivicConnect could capture and track a maintenance request, but the system itself does not execute or manage the actual maintenance action, except for the information required to manage the life cycle of the maintenance request.

1.3.4 Deferred and Future Scope

Some capabilities might have stakeholder value in the future, but the stakeholder requirements, evidence or engineering analysis are not yet available in the initial scope and are intentionally not included.

Future/optional features may include:

- Interoperability with external communications or notifications; and
- Utilized advanced analytics or management dashboards that provided more than the minimum report information; and
- The ability to integrate with other organisational systems;
- mobile-specific applications;
- Automated request assignment/prioritisation;
- expanded notification mechanisms;
- Additional reporting or data-export functionality; and
- Enhancements requested by other stakeholders in the project later in its activities.

These items are suggestions, not requirements. Evidence of stakeholder value and impact on scope, schedule, resources, quality, security and risk would be required for their inclusion.

Also, the final tech stack, architecture, detailed design of the DB, detailed UI implementation, API implementation and production deployment are not considered to be accepted M1 scope decisions. The M1 brief explicitly identifies these as places where it is not necessary to make early decisions on engineering now.

1.3.5 Deliberate Scope Exclusions and Deferments

The team has intentionally eliminated or postponed functionality that has not been essential to show the fundamental value of CivicConnect to the stakeholder. This isn't because the added features are not worth it, but because of scope discipline.

One of the most critical planned deferments is integration with external communication platforms like WhatsApp. The fragmented handling of requests is one of the ways WhatsApp is impacting business processes, but it is not something CivicConnect must deal with explicitly given that WhatsApp is not a platform that it must integrate with. To create a controlled life cycle of a central service request is a primary requirement.

Incorporating such an integration would add other requirements with respect to external APIs, authentication, security, privacy, reliability, testing and maybe even operational cost. It would, therefore, expand the engineering responsibilities with no need to prove the concept of centralised request submission, tracking and management.

It's the same with advanced analytics, automation and additional optional features. The Master Project Brief clearly states that each and every extra feature gives rise to obligations to specify, design, secure, implement, test, document, deploy and maintain it, and acknowledges that a smaller controlled solution may turn out to be more impressive Software Engineering than a larger unfinished solution.

Therefore, these capabilities are postponed until the need of the stakeholders and engineering evidence are adequate. This helps to safeguard the scope from uncontrolled expansion, while also allowing for scope to be evolved in the future.

In dynamic software environments, with multiple stakeholders, (Silva, et al., 2024) offer relevant support for this approach by studying the volatility of requirements. They emphasize the need to proactively adapt to changing needs and not to let changes creep into a project without being considered. The new stakeholder request should then be subjected to an analysis of the impacted requirements, schedule, resources, quality, security, and risk prior to altering the scope baseline for CivicConnect.

1.4 Project Constraints

CivicConnect needs to be designed under a variety of interdependent constraints that help shape the team's expectations and expectations of how CivicConnect will be developed. The main constraints identified for the project are scope, schedule, cost/resources, quality and security. These are not to be taken as individual items – each decision in relation to one constraint can have an impact on another.

In software projects, time pressure is an important factor. (Gilal, et al., 2023), performed a review of software-development literature, and concluded that time pressure can create negative project outcomes, such as poorer software quality and higher project risk. In CivicConnect, the schedule pressure must therefore be controlled through scope and prioritisation - not by just eliminating engineering and quality work.

1.4.1 Scope Constraints

The scope must be kept on track and under control throughout the project, as the commitment of CivicConnect requires. The system must offer the minimum business capabilities needed to submit requests, to manage requests and to oversee the management and it is mandatory to include further functionality if it is valuable to the stakeholders and its impact on the project is evaluated against the wider project impact.

So, the primary scope constraint is the amount of additional functionality the team can take on after the baseline is in place. New stakeholder requests are not automatically committed

requirements. Changes to anything must be assessed before they are made, to see if they will affect the existing requirements, schedule, resources, quality, security and risk.

This constraint will allow for the intentional exclusions and deferments of the scope baseline. The idea of reducing scope is to keep the team from agreeing to do more than can be designed and tested with the current project constraints, but it is not a rejection of potentially valuable functionality.

1.4.2 Schedule Constraints

There are four formal milestones for CivicConnect during the SEN381 delivery period. Milestones are dependent on one another, and any failures or slowdowns to one milestone may have consequences downstream on subsequent milestones.

The schedule thus restricts what can be achieved in terms of functionality and engineering by the three-person team. In addition to being technically feasible, scope decisions need to consider if there is enough time to be able to specify, design, secure, test, document and defend the feature.

Avoid using workarounds for engineering controls when dealing with schedule pressures. Gilal, Omar and Rejab (2023) link time pressure in software development with several negative consequences such as poor software quality and high project risk. It is thus preferable that CivicConnect have a controlled scope and prioritise essential value for stakeholders than to add functionality and must cut testing, security or documentation efforts to time.

1.4.3 Cost and Resource Constraints

Developers of CivicConnect are a small team of 3 students, presenting a challenge in this area. The development, research, documentation, testing and project-management effort can be split among these three team members, and each member of the team must be responsible for knowing the whole project.

The project has a cost constraint also. The Master Project Brief asks the team to choose services that are either free or low cost, where possible, and to be aware of the limits of their services and the costs they are likely to incur outside of the educational setting.

Therefore, when looking at hosting, development services, infrastructure and other technologies in the future, cost should be taken into consideration. But it is not always the cheapest solution that is used, if it results in unacceptable restrictions on Quality, Security, Maintainability or deployment.

The other way around, the existing knowledge and available time of a team are resources as well. A technology or solution that will need significant additional learning time will take up time that otherwise could be devoted to implementation, testing and verification. Purchasing and operating and maintaining the chosen technology should be considered in subsequent technology decisions, not only for financial costs but also for human resources.

1.4.4 Quality Constraints

CivicConnect must meet specific quality standards and not be judged successful just because it has featured that work. The Master Project Brief must include quality attributes, which are then further substantiated by quantitative evidence such as non-functional requirements, acceptance criteria, testing evidence and quality gates.

This puts a restriction on scope and implementation. With each committed feature, there is an expectation that the feature needs to be implemented and that it needs to satisfy the functional and quality requirements of that feature. If the schedule and/or resources remain the same, then a larger scope will be harder to cover because there's less time and resources.

Quality therefore cannot be assessed as a final test but must be conceived from the outset of the development process. In the case of schedule pressure, decreasing the amount of testing or loosening the acceptance criteria to match the speed would not get rid of the quality constraint, it would just add uncertainty and increase project risk.

1.4.5 Security Constraints

Security is not something that can be tacked on to CivicConnect after it has been implemented; it is a constraint throughout the entire lifetime of the CivicConnect. Security needs to impact requirements, repository controls, secrets handling, threat review, testing/scanning and deployment controls in the Master Project Brief.

This is especially critical, since requests for service could include information that is not meant to be displayed to all users. The system must then be able to differentiate between the information and the actions that may be requested by the requester, staff and management according to the appropriate authorisation.

Other desired stakeholder functionality may be limited by security requirements. For instance, some of the comments or information in the flow may be restricted to authorized users, whereas the requester may need to see everything that is done with their request. It is important, therefore, to ensure greater transparency while providing adequate access to information.

Project resources are also used on security. Engineering effort is required for all of authentication, authorisation, secure handling of information, testing and repository controls. So, it's not a choice when the schedule or resources become tight, the team cannot leave security work aside.

1.4.6 Constraint Interactions and Trade-offs

Each of the CivicConnect constraints has an impact on the other, so altering one constraint can affect multiple constraints.

A good example is when an extra feature is requested at a late stage. When a Stakeholder wants a major new feature added when the scope is baseline and the team size and project deadline do not change, the following interaction takes place:

As the scope of work increases, the workload increases, the schedule pressure increases, the amount of time spent on quality/security effort decreases, and the risk increases.

Having to accept the feature without changing another constraint might allow less time to be spent on requirements analysis, implementation, testing, security verification, and documentation. The relationship was supported by the relevant evidence found in the works of (Gilal, et al., 2023) who in their review found that the consequences of time pressure in software-development projects include compromised quality and increased risk.

The engineering desired solution would, therefore, not be to automatically take on the extra functionality. The team would first assess the value of the stakeholders, the requirements that

would be impacted, the implementation effort and the impact to the schedule, resources, quality, security and current risks. This could then be accepted, reduced, deferred or rejected as evidence for the feature.

Another trade-off is in making cost decisions. A free service could be used to keep project within financial boundaries, but lack of reliability, capacity, security or support might impact quality or future deployment. On the other hand, an able paying service may be able to enhance some quality attributes and cause an unnecessary money drain. The Master Brief must therefore be based on requirements/constraints and not just a stack/Platform.

The team will then view the constraints as a system of interconnected engineering constraints instead of engineering constraints that are treated individually. If, because of a significant decision and/or change, one constraint is impacted, impact on the other constraints should be considered before committing to it.

1.5 Engineering Decision Log

1.5.1 Purpose of the Engineering Decision Log

The Engineering Decision Log is a structured record of key engineering decisions that were made for the CivicConnect project. It is not just about documenting the outcome of the team's decision, but also the context in which it was made, the constraints and alternatives faced, the logic for the chosen approach, the trade-offs and risks involved, and the potential downstream impacts.

It is critical to keep these in hand because the evidence presented will be used in engineering decisions at a certain moment in the project. As CivicConnect evolves, the team should consider the rationale behind an earlier decision, understand if its assumptions are still accurate, and/or consider how a different option would impact.

(Demir, et al., 2024) conducted an industrial survey of 101 software practitioners to study the practices in the architectural decision-making process. Their study revealed that generally practitioners record architectural decisions, and that there is a need for improvement in information sharing, decision keeping and the rationale for decisions. This helps to maintain a controlled decision record that will ensure that not only the result of the decision is recorded but the reasoning as well.

The following structure will thus be used for the significant decisions for CivicConnect:

Decision ID → Context → Constraints → Alternatives → Decision → Rationale → Trade-offs → Risks → Evidence → Later Consequence

As more evidence is gathered, and more engineering decisions are needed, the Engineering Decision Log will develop throughout the project.

1.5.2 Milestone 1 Engineering Decisions

Only decisions that are justified by the evidence available during Milestone 1 are recorded as confirmed decisions. Architecture, technology and implementation decisions that have not yet been sufficiently evaluated are deliberately excluded from the confirmed M1 decisions.

<i>ID</i>	<i>Decision</i>	<i>Context and Alternatives</i>	<i>Rationale</i>	<i>Trade-off/Risk</i>	<i>Evidence/Later Consequence</i>
ED-01	Establish a controlled CivicConnect scope baseline focused on the minimum business capabilities.	The team could commit only to the core capabilities or expand the baseline with additional desirable functionality.	A controlled core scope directly addresses the identified business need while remaining more achievable within the three-person team and project schedule.	Some potentially valuable functionality will not initially be included. This reduces scope growth but may require future extension.	The baseline will guide the requirements and RTM. Any later scope addition must be assessed before becoming committed work.
ED-02	Do not include direct WhatsApp integration in the initial committed scope.	The existing process uses WhatsApp as one of several fragmented channels. Alternatives included direct WhatsApp integration or establishing CivicConnect itself as the controlled request-management platform.	The underlying problem is fragmented request management rather than the absence of WhatsApp integration. Direct integration is not required to demonstrate the core stakeholder value of centralised request submission, tracking and management.	Users familiar with WhatsApp will not initially be able to manage requests through that channel. However, integration complexity, external dependency, security and additional testing obligations are avoided.	WhatsApp integration remains possible future scope if stakeholder value and later technical evidence justify it.
ED-03	Use stakeholder needs as an upstream source for CivicConnect	Requirements could be recorded independently or explicitly linked to the stakeholder	Linking requirements to stakeholder needs provides evidence that	Maintaining traceability requires additional documentation effort but reduces the	Stakeholder → Need → Requirement → Acceptance Criteria.

	requirements and acceptance criteria.	needs from which they originate.	committed functionality contributes to the identified business problem and stakeholder value.	risk of unsupported or unnecessary requirements.	
ED-04	Treat additional stakeholder feature requests as proposed scope changes rather than automatically accepting them.	New requests could be added informally or assessed before changing the baseline.	Automatic inclusion could create uncontrolled scope growth and downstream effects on schedule, resources, quality, security and risk.	Some stakeholder requests may be delayed or rejected, but the team retains control over project commitments.	Future proposed changes should trigger impact analysis and updates to affected requirements and traceability artefacts if approved.

These decisions establish controls rather than prematurely selecting technical solutions. They provide a foundation for the later requirements, risk, architecture and implementation activities while remaining appropriate to the purpose of Milestone 1.

1.5.3 Deliberately Deferred Decisions

Several significant engineering decisions have deliberately **not** been made during Milestone 1 because the evidence required to make defensible choices is not yet sufficient. Deferring these decisions is intentional and prevents the team from selecting solutions before the relevant requirements, constraints and alternatives have been adequately evaluated.

<i>Deferred Decision</i>	<i>Why It Is Deferred</i>	<i>Evidence Needed Before Decision</i>	<i>Risk of Deciding Too Early</i>
Final technology stack	M1 establishes the engineering and requirements baseline rather than committing to implementation technologies.	Baselined FRs/NFRs, team capability, security requirements, cost, deployment needs and comparative technology evidence.	A preferred technology could be selected because of familiarity or popularity rather than project suitability.
Final software architecture	Architecture should respond to established requirements and quality attributes.	Confirmed requirements, quality priorities, expected workload, security needs,	Premature architecture could create unnecessary complexity or make

		maintainability requirements and architectural alternatives.	later requirements harder to satisfy.
Detailed database and persistence design	The information that must be stored and its relationships must first be derived from the requirements.	Confirmed data requirements, lifecycle states, audit/traceability needs, security requirements and later architectural decisions.	Early schema decisions may embed incorrect assumptions and become expensive to change.
Detailed UI design	Stakeholder capabilities have been identified, but detailed interaction and usability decisions require further requirements/design work.	Confirmed workflows, acceptance criteria, usability requirements and stakeholder feedback.	Premature screens may drive the requirements instead of being designed from them.
Deployment platform and production infrastructure	Operational requirements and later technology decisions have not yet been sufficiently evaluated.	Technology stack, expected workload, cost limitations, security requirements, availability needs and deployment alternatives.	Early platform commitment may create unnecessary cost, technical limitations or migration work.
External integrations	Integrations such as WhatsApp or other external services are not necessary for the initial core scope, and their value has not yet justified their complexity.	Confirmed stakeholder need, API/platform constraints, security/privacy implications, operational cost and implementation effort.	External dependencies could unnecessarily increase scope, security exposure, cost and schedule pressure.

2. REQUIREMENTS BASELINE

Functional Requirements (FRs):

Table 1: Functional Requirements for CivicConnect.

ID	FR	Source/Stakeholder	Priority
----	----	--------------------	----------

FR-001	System shall allow requesters to submit service requests	Requesters	Should
FR-002	Requester shall be able to view request status	Requesters	Should
FR-003	Requester shall be able to view request history	Requestors	Should
FR-004	Requester shall receive feedback when request is accepted/rejected/updated/completed	Requestors	Should
FR-005	The system shall allow authorised staff to view service requests relevant to their responsibilities	Staff	Must
FR-006	The system shall allow authorised staff to search, filter, and sort service requests using useful request info	Staff	Must
FR-007	The system shall allow authorised staff to assign or accept responsibility for a service request	Staff	Must
FR-008	The system shall allow authorised staff to update the status of a service request using controlled status transitions	Staff	Must
FR-009	The system shall allow authorised staff to record relevant actions, comments and resolution info against a service request	Staff	Must
FR-010	The system shall allow appropriately authorised staff to resolve or close a service request	Staff	Must
FR-011	The system shall provide management with up-to-date service activity info relevant to operational oversight	Management/Oversight	Must
FR-012	The system shall allow authorised management users to identify open, overdue, resolved and closed service requests	Management/Oversight	Must
FR-013	The system shall provide info that supports accountability and service-performance analysis	Management/Oversight	Must
FR-014	The system shall maintain a controlled lifecycle record for each service request, including its relevant status changes, responsibility, recorded actions and resolution/closure info for the sake of avoiding duplicate request requirements	Community-focused organisation	Must

Source: (Klasman, 2024), (Hartatik, 2022)

Legend	
Must	Minimum essential required priority, a need

Should	Important priority but not as essential as 'Must'
Could	Beneficial extra functionality that should only be included if scope, schedule, cost, quality, and risk allow

Non-Functional Requirements (NFRs):

Table 2: Non-Functional Requirements for CivicConnect.

ID	NFR Category	NFR	Measure/Acceptance Target	Source	Priority
NFR-001	Security/Authorisation	The system shall restrict access to service-request info and functions according to the user's authorised responsibilities	Users will be unable to access functions or request info outside their authorised permissions, an "access denied" message will be seen. 100% of requests lacking valid JSON Web Tokens (JWT) or session permissions must be rejected instantly (unauthorized request rejection rate).	Requesters/ Staff/ Management/Oversight	Must
NFR-002	Security/Privacy	The system shall protect sensitive service-request info from unauthorised access or disclosure	Security verification shall demo that unauthorised users cannot access protected request info, an "unauthorised user" message will be seen. All admin logins require multi-factor authentication (MFA).	Requesters/ Staff	Must
NFR-003	Security/Least Privilege	The system shall apply least-privilege principles to user access.	Each role will only have the permissions necessary for its approved	Requesters/ Staff/ Management/Oversight	Must

			responsibilities, a “action is not permitted” message will be seen.		
NFR-004	Reliability/ Data Integrity	The system shall maintain service-request info consistently as well as also not accept incomplete/invalid request info where it is needed to process a request.	Make sure database shows 100% roll back when a datatype mismatch error is triggered during a status update	Requesters	Must
NFR-005	Usability	The system shall provide a usable process for submitting and monitoring service requests.	Submitting a request must take a user <60 seconds. Locating the status/history page must take <30 seconds. From the main dashboard, a user must be able to reach the request submission form in <= 2 clicks, and find their request history in <= 2 clicks	Requesters	Must
NFR-006	Usability	Request status and important request info shall be presented in a manner that is understandable to the intended user.	100% of volunteer testers can correctly read and identify the ticket's current status, owner, and resolution details within 5 seconds without help. User satisfaction score must be average of 4/5	Requesters	Must
NFR-007	Performance	The system shall provide responsive interaction for normal service-	Under normal expected project conditions, 95% of standard request-management operations should	Requesters/Staff/ Management/Oversight	Should

		request operations	complete within 3 seconds		
NFR-008	Maintainability	The solution shall be structured and documented sufficiently to allow controlled future modification w/o requiring unnecessary rework	A live RTM will be maintained. When a requirement is changed, the requirement ID will be mentioned in the Git commit message. (i.e. git commit -m "Fix: updated time formatting to UTC to satisfy NFR-004)	Staff/ Management/Oversight	Should

Source: (Hartatik, 2022), (Ray, 2026), (Ibrahim & Ta'a, 2015)

Acceptance Criteria:

Table 3: Functional Acceptance Criteria.

AC ID	Requirement	Acceptance Criteria
FR-AC-001	FR-001	Given a requester has valid information for a service issue, when they submit the request, then the system shall create a service request that can subsequently be identified and managed.
FR-AC-002	FR-001	Given required request information is missing or invalid, when the requester attempts to submit the request, then the system shall prevent submission and identify the information that needs correction.
FR-AC-003	FR-002	Given a requester is submitting a service request, when they select a category, then the category shall come from the controlled category mechanism rather than unrestricted category input.
FR-AC-004	FR-002	Given a requester has submitted a service request, when they view their request, then the current request status shall be displayed.
FR-AC-005	FR-003	Given a requester has previously submitted

		requests, when they access their request history, then the system shall display their previously submitted requests.
FR-AC-006	FR-004	Given a request undergoes an accepted, rejected, updated or completed outcome, when that event is recorded, then meaningful feedback shall be made available to the requester.
FR-AC-007	FR-005	Given a staff member is authorised to view particular service requests, when they access the request-management area, then they shall be able to view the requests relevant to their responsibilities.
FR-AC-008	FR-006	Given multiple service requests exist, when an authorised staff member searches, filters or sorts them, then the system shall return or arrange requests according to the selected useful criteria.
FR-AC-009	FR-006	Given an authorised staff member selects a service request, when they open it, then the full available request details shall be displayed.
FR-AC-010	FR-007	Given a service request requires ownership, when an appropriately authorised staff member assigns or accepts responsibility, then the request shall reflect the responsible staff member.
FR-AC-011	FR-008	Given an authorised staff member manages a request, when they attempt to change its status, then only permitted status transitions shall be allowed.
FR-AC-012	FR-008	Given a status transition is not permitted, when a staff member attempts that transition, then the system

		shall prevent the invalid transition.
FR-AC-013	FR-009	Given an authorised staff member takes relevant action on a request, when they record the action, comment or resolution information, then the information shall become associated with the relevant request.
FR-AC-014	FR-010	Given a service request meets the conditions for resolution or closure, when an appropriately authorised staff member resolves or closes it, then its lifecycle state shall reflect the authorised resolution or closure.
FR-AC-015	FR-011	Given management has authorised oversight access, when management accesses service activity information, then useful information about service activity shall be available.
FR-AC-016	FR-012	Given service requests exist in different lifecycle states, when management views request oversight information, then open, overdue, resolved and closed requests shall be identifiable.
FR-AC-017	FR-012	Given service requests exist across different categories or statuses, when management selects a supported dimension, then relevant request information shall be viewable according to that dimension.
FR-AC-018	FR-013	Given management needs to assess service performance, when they access the relevant information, then the information available shall

		support accountability and service-performance analysis.
FR-AC-019	FR-014	Given a service request is submitted, when authorised users subsequently manage it, then its relevant status, responsibility, actions and resolution/closure information shall remain associated with the request lifecycle.
FR-AC-020	FR-014	Given the lifecycle of a request has changed, when an authorised user reviews the request, then sufficient lifecycle information shall be available to determine what happened to the request.

Table 4: Non-Functional Acceptance Criteria.

AC ID	Requirement	Acceptance Criteria
NFR-AC-001	NFR-001	Given users with different authorised responsibilities, when each attempts to access functions or request information, then access shall be permitted or denied according to their authorised permissions.
NFR-AC-002	NFR-002	Given a request contains information that is not intended for an unauthorised user, when that user attempts to access it, then the information shall not be disclosed.
NFR-AC-003	NFR-003	Given a defined user role, when its permissions are tested, then the role shall have only the permissions required for its approved responsibilities.
NFR-AC-004	NFR-004	Given a service request has been created, when its status, responsibility or resolution information

		changes, then previously recorded request information shall remain consistent.
NFR-AC-005	NFR-004	Given required information is absent or invalid, when the request is submitted, then the system shall reject the invalid submission and identify the problem.
NFR-AC-006	NFR-005	Given a representative requester is unfamiliar with the system, when they attempt to submit and monitor a service request, then they shall be able to complete these tasks without requiring technical knowledge.
NFR-AC-007	NFR-006	Given a requester views an existing request, when they review its information, then they shall be able to correctly identify its current status and relevant information.
NFR-AC-008	NFR-007	Given the system is operating under agreed normal project conditions, when standard request-management operations are measured, then at least 95% shall complete within the agreed 3-second target.
NFR-AC-009	NFR-008	Given an approved requirement changes in a later milestone, when the change is processed, then the affected requirement, design/implementation artefacts and verification evidence shall be identifiable through project traceability.

Requirements Traceability Matrix (RTM):

Table 5: Requirements Traceability Matrix for CivicConnect.

Req ID	Requirement	Source	Priority	Implementation/Work	Verification	Status
FR-001	System shall allow requesters to submit service requests	Requesters	Should	Not yet created	FR-AC-001, FR-AC-002	Baselined
FR-002	Requester shall be able to view request status	Requesters	Should	Not yet created	FR-AC-003, FR-AC-004	Baselined
FR-003	Requester shall be able to view request history	Requesters	Should	Not yet created	FR-AC-005	Baselined
FR-004	Requester shall receive feedback when request is accepted/rejected/updated/completed	Requesters	Should	Not yet created	FR-AC-006	Baselined
FR-005	The system shall allow authorised staff to view service requests relevant to their responsibilities	Staff	Must	Not yet created	FR-AC-007	Baselined
FR-006	The system shall allow authorised staff to search, filter, and sort service requests using useful request info	Staff	Must	Not yet created	FR-AC-008, FR-AC-009	Baselined
FR-007	The system shall allow authorised staff to assign or accept responsibility for a service request	Staff	Must	Not yet created	FR-AC-010	Baselined
FR-008	The system shall allow authorised staff to update the status of a service request using controlled status transitions	Staff	Must	Not yet created	FR-AC-011, FR-AC-012	Baselined
FR-009	The system shall allow authorised staff to record relevant actions, comments	Staff	Must	Not yet created	FR-AC-013	Baselined

	and resolution info against a service request					
FR-010	The system shall allow appropriately authorised staff to resolve or close a service request	Staff	Must	Not yet created	FR-AC-014	Baselined
FR-011	The system shall provide management with up-to-date service activity info relevant to operational oversight	Management/Oversight	Must	Not yet created	FR-AC-015	Baselined
FR-012	The system shall allow authorised management users to identify open, overdue, resolved and closed service requests	Management/Oversight	Must	Not yet created	FR-AC-016, FR-AC-017	Baselined
FR-013	The system shall provide info that supports accountability and service-performance analysis	Management/Oversight	Must	Not yet created	FR-AC-018	Baselined
FR-014	The system shall maintain a controlled lifecycle record for each service request, including its relevant status changes, responsibility, recorded actions and resolution/closure info for the sake of avoiding duplicate request requirements	Community-focused organisation	Must	Not yet created	FR-AC-019, FR-AC-020	Baselined
NFR-001	The system shall restrict access to service-request info and functions according to the user's authorised responsibilities	Requesters/ Staff/ Management/Oversight	Must	Not yet created	NFR-AC-001	Baselined
NFR-002	The system shall protect sensitive service-request info from unauthorised access or disclosure	Requesters/ Staff	Must	Not yet created	NFR-AC-002	Baselined
NFR-003	The system shall apply least-privilege principles to user access.	Requesters/ Staff/ Management/Oversight	Must	Not yet created	NFR-AC-003	Baselined

NFR-004	The system shall maintain service-request info consistently as well as also not accept incomplete/invalid request info where it is needed to process a request.	Requesters	Must	Not yet created	NFR-AC-004, NFR-AC-005	Baselined
NFR-005	The system shall provide a usable process for submitting and monitoring service requests.	Requesters	Must	Not yet created	NFR-AC-006	Baselined
NFR-006	Request status and important request info shall be presented in a manner that is understandable to the intended user.	Requesters	Must	Not yet created	NFR-AC-007	Baselined
NFR-007	The system shall provide responsive interaction for normal service-request operations	Requesters/ Staff/ Management/Oversight	Should	Not yet created	NFR-AC-008	Baselined
NFR-008	The solution shall be structured and documented sufficiently to allow controlled future modification w/o requiring unnecessary rework	Staff/ Management/Oversight	Should	Not yet created	NFR-AC-009	Baselined

Source: (Belgium Campus iTversity, 2026)

Traceability Example:

Following a single requirement (Request Status Visibility) through the project lifecycle:

Stakeholder need

Requester needs visibility of what is happening with a submitted service request.

↓

Problem evidence

The current process gives requesters limited visibility of whether a request has been received, assigned, pending, solved or closed.

↓

Requirement

FR-002: The system shall allow a requester to view request status.

↓

Acceptance criterion

FR-AC-004: Given a requester has submitted a service request, when they view their request, then the current request status shall be displayed.

↓

Future design

TBC in M2

The architecture/design team will determine how this requirement is represented in the solution.

↓

Future implementation

TBC in M3

Implementation evidence will be linked to FR-002.

↓

Future test

TBC in M3

A system/functional test will verify FR-AC-004.

↓

Future acceptance/release evidence

TBC in Final Milestone Submission

Stakeholder validation will demonstrate whether the requirement actually satisfies the original requester need.

This demonstrates why the RTM is more than a spreadsheet: it provides a controlled connection between the original stakeholder problem and later engineering evidence.

3. ENGINEERING CONTROL & FORWARD PLANNING

Table 1: Risk Register

Introduction:

The initial Risk Register identifies and evaluates risks that may affect the successful engineering and delivery of CivicConnect. Risk management is performed continuously throughout the software lifecycle so that threats can be identified, analysed and treated before they negatively affect project objectives (ISO/IEC/IEEE, 2021). The register records each risk together with its probability, impact, priority, mitigation strategy, contingency action, owner and current status.

ID	Risk	Probability	Impact	Priority	Mitigation	Contingency	Owner	Status
R-01	Loss or corruption of CivicConnect data may affect data integrity and service continuity.	Medium	High	High	Define backup, recovery and transactional integrity requirements before database technology is finalised.	Restore data from a verified backup and investigate the cause of corruption before normal operation resumes.	Megan Wathen	Open
R-02	Failure or unavailability of external or internal service interfaces may prevent residents or administrators	Medium	High	High	Identify service dependencies and define availability, error-handling and recovery requirements.	Provide controlled failure handling and restore/reconnect affected services when available.	Megan Wathen	Open

	from completing system functions.							
R-03	Weak authentication, token or session management could allow unauthorised access to CivicConnect functions or information.	Medium	Very High	High	Define secure authentication, authorisation and session-management requirements before implementation decisions are made.	Revoke compromised credentials/sessions, investigate the incident and apply corrective access controls.	Megan Wathen	Open
R-04	Failures or abnormal system behaviour may not be detected quickly enough if monitoring and logging requirements are inadequate.	Medium	Medium/High	Medium/High	Define logging, monitoring and alerting requirements before deployment decisions are made.	Investigate logs and system evidence, isolate the affected component and perform recovery procedures.	Megan Wathen	Open

Table 2 : CivicConnect Forward Engineering Considerations Register

Concern	Why it matters now	Future decision affected	Information still needed	Risk if ignored
Security	CivicConnect may process resident and municipal information that must be protected against unauthorised access or disclosure.	Authentication, authorisation, encryption and security controls.	Threat model, applicable regulatory requirements and identity-management requirements.	Data exposure, unauthorised access, loss of stakeholder trust and possible compliance consequences.
Testability	Requirements and system functions must be capable of being verified during later development and acceptance testing.	Test strategy, environment separation and automated testing approach.	Required test coverage, available testing tools, test data and acceptance requirements.	Defects may remain undetected and requirements may be difficult to verify.
Continuous Integration / Automated Builds	Future development may involve frequent changes from multiple team members that need to be integrated reliably.	CI/CD tooling, build automation and branching workflow.	Repository integration requirements, build process and dependency-management needs.	Integration problems, inconsistent builds and increased manual errors.
Deployment Environment	Deployment requirements can influence later architecture, security, availability and operational decisions.	Hosting approach and DEV/TEST/PROD environment strategy.	Infrastructure constraints, municipal policies, expected availability and operational requirements.	An unsuitable deployment approach may result in security, reliability or operational problems.
Backup, Recovery & Data Migration	CivicConnect data may need to be restored following data loss, corruption or service failure.	Backup strategy, disaster recovery and future migration approach.	Required recovery time, acceptable data-loss tolerance and existing data sources.	Permanent data loss or extended service interruption.

Scalability / Performance	CivicConnect should remain responsive if user demand increases or varies over time.	Architecture, hosting capacity and performance strategy.	Expected user numbers, peak workloads and measurable performance requirements.	Slow response times, service disruption and poor user experience.
Maintainability / Observability	The system will need to be maintained, monitored and diagnosed after implementation.	Logging, monitoring, alerting and maintainability approach.	Required operational metrics, alert thresholds, maintenance responsibilities and support capacity.	Failures may remain undetected, troubleshooting may take longer and maintenance effort may increase.

GitHub and Team Governance

GitHub is used as the primary version-control and collaboration environment for the CivicConnect project. The repository is intended not only to store project artefacts, but also to provide evidence of controlled engineering activity throughout the project. Significant changes to requirements, registers, documentation and later implementation artefacts are managed through issues, dedicated branches, commits, Pull Requests and peer review.

The `main` branch represents the controlled project baseline. Substantive changes should therefore not be introduced directly into `main`. Instead, work is performed on a separate branch and submitted through a Pull Request for review before being merged. This provides a visible history of who proposed a change, what was changed, how it was reviewed and when it became part of the controlled baseline.

Table 3: CivicConnect GitHub Governance Controls

Governance Control	How CivicConnect Uses It	Purpose
Protected main branch	The main branch is protected from uncontrolled direct changes.	Protects the controlled project baseline.
Issues / Tasks	Significant project activities are recorded as GitHub Issues and assigned to responsible team members.	Provides task ownership and traceability.
Separate branches	Significant work is completed on dedicated branches before being proposed for merge.	Keeps unfinished work separate from the controlled baseline.

Meaningful commits	Commit messages describe the engineering change being made.	Creates an understandable history of project development.
Pull Requests	Significant changes are submitted through Pull Requests before being merged into main.	Creates a formal review point for changes.
Peer review	Team members review the proposed change and may request corrections before approval.	Helps identify mistakes and ensures shared accountability.
Two-person approval	Substantive changes are reviewed by the other two members of the three-person team before merging, where required by the project governance rules.	Prevents the author from being the only person validating their own work.
Review comments	Review feedback and responses are retained in the Pull Request history.	Provides evidence of genuine engineering discussion and correction.
Controlled merge	Changes are merged only after required reviews and outstanding discussions have been resolved.	Ensures only reviewed work enters the baseline.

GitHub Evidence

Evidence	Status	Location
Repository created	Complete	GitHub repository
Team members added	Complete	Repository access settings
main branch protection	Complete	Repository rules/settings
Risk Register Issue	Complete	GitHub Issues
Risk Register branch	Complete	Repository branches
Risk Register Pull Request	Complete	Pull Requests
Two peer approvals	Pending	Pull Request review history
Forward Engineering Register PR	Pending	Pull Requests
AI Usage Register PR	Pending	Pull Requests

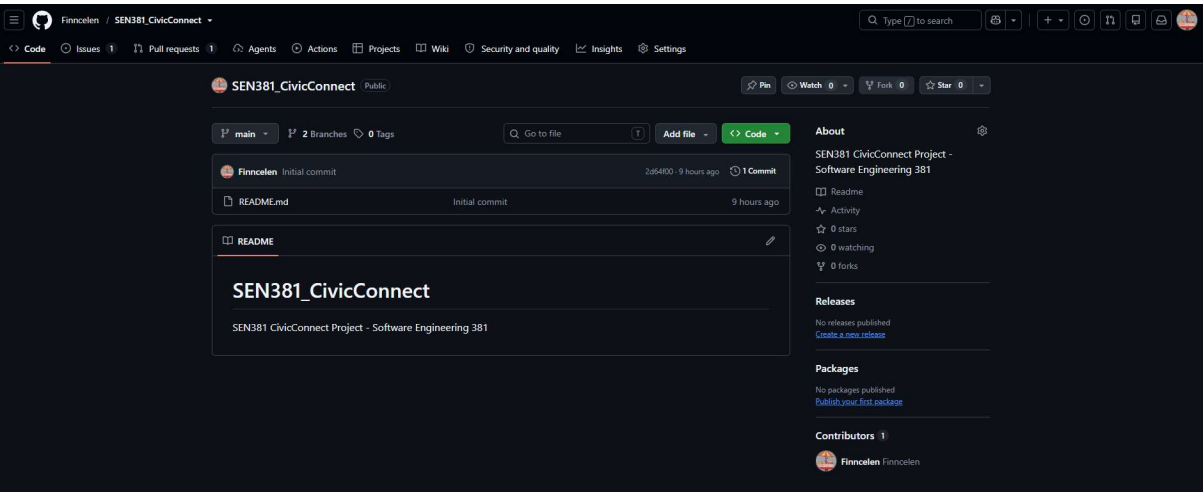
Controlled Change Flow

Issue → Branch → Commit → Pull Request → Peer Review → Approval → Merge to main

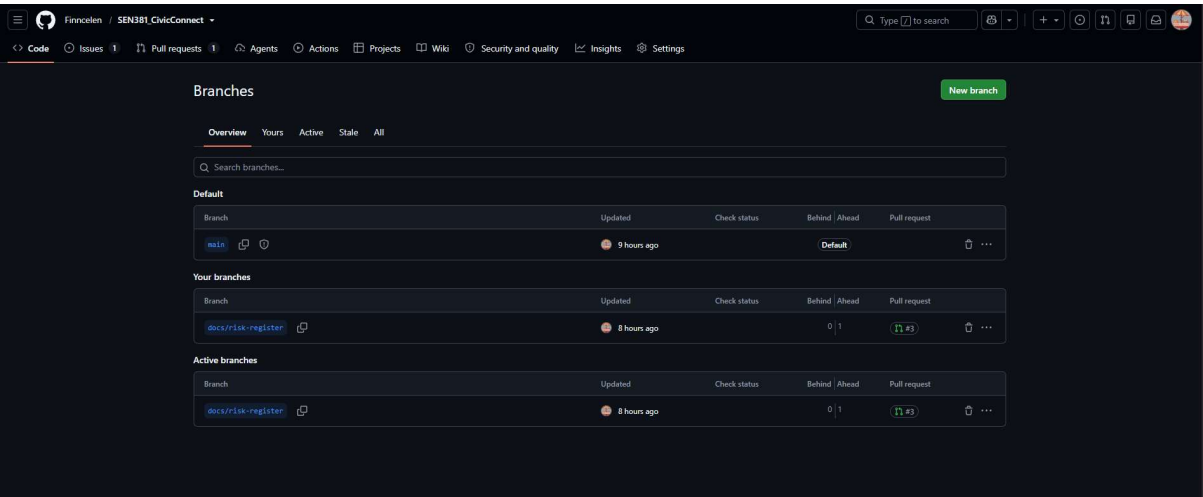
This workflow is used to ensure that significant project changes are planned, traceable, reviewed and approved before they become part of the controlled CivicConnect baseline. Issues record the work to be completed, branches isolate changes from main, commits provide a history of modifications, and Pull Requests provide a formal point for peer review and approval. Once the required reviews are completed and any identified concerns are resolved, the change may be merged into main.

Evidence Screenshots

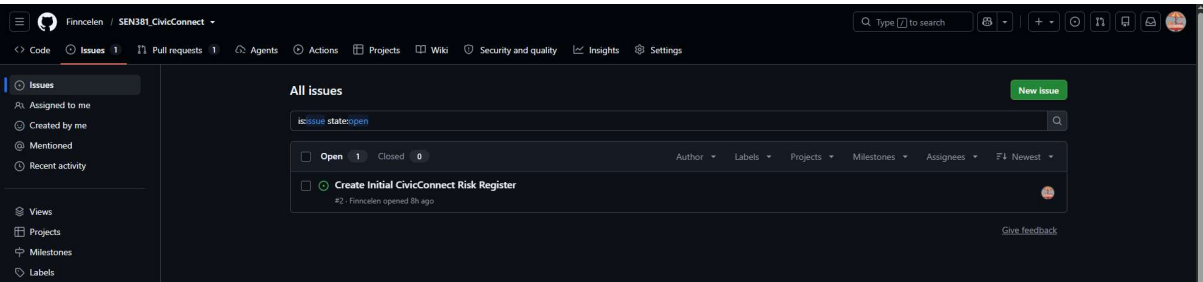
Repository Made



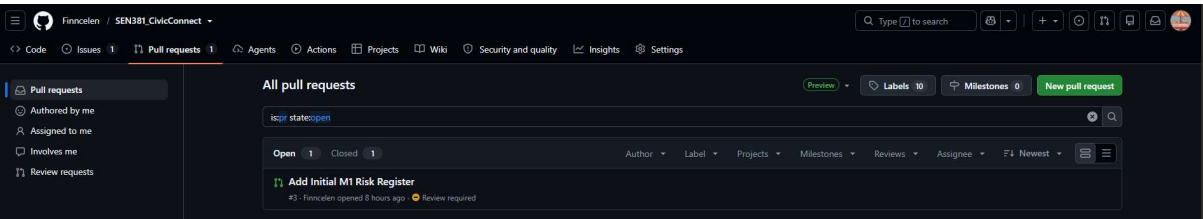
Risk register branch



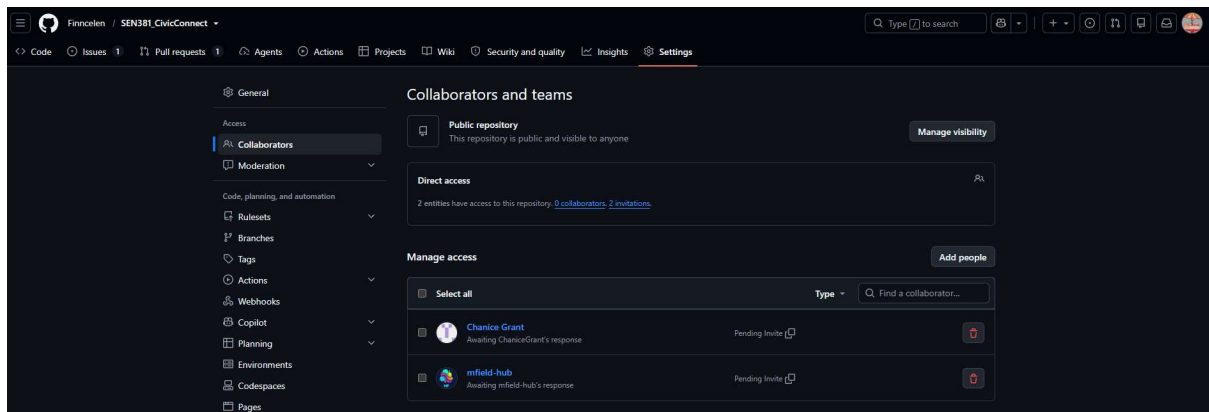
Issue made



Pull Request made



All team members invited



Protect main Ruleset made

Finncelen / SEN381_CivilConnect

Type to search

CodeIssues1Pull requests1AgentsActionsProjectsWikiSecurity and qualityInsightsSettings

General

Access

Collaborators

Moderation

Code, planning, and automation

Rulesets

Rulesets

Rulesets

New ruleset

All

Protect Main
3 branch rules • targeting 1 branch

Rulesets / Protect Main

Active

...

Ruleset Name *

Protect Main

Enforcement status

Active

Bypass list

Exempt roles, teams, agents, apps, and users from this ruleset by adding them to the bypass list.

Actors

AllRolesTeamsAppsUsersOthers

+ Add bypass

Bypass list is empty

Target branches

Which branches should be matched?

Branch targeting criteria

Add target

+ Default

Applies to 1 target: main

Rules
Which rules should be applied?

Branch rules

☐ Restrict creations
Only allow users with bypass permission to create matching refs.

☐ Restrict updates
Only allow users with bypass permission to update matching refs.

☒ Restrict deletions
Only allow users with bypass permissions to delete matching refs.

☐ Require linear history
Prevent merge commits from being pushed to matching refs.

☐ Require deployments to succeed
Choose which environments must be successfully deployed to before refs can be pushed into a ref that matches this rule.

☐ Require signed commits
Commits pushed to matching refs must have verified signatures.

☒ Require a pull request before merging
Require all commits be made to a non-target branch and submitted via a pull request before they can be merged.

Show additional settings ▾

☐ Require status checks to pass
Choose which status checks must pass before the ref is updated. When enabled, commits must first be pushed to another ref where the checks pass.

☒ Block force pushes
Prevent users with push access from force pushing to refs.

☐ Require code scanning results
Choose which tools must provide code scanning results before the reference is updated. When configured, code scanning must be enabled and have results for both the commit and the reference being updated.

☐ Require code quality results
Choose which severity levels of code quality results should block pull request merges. When configured, a code quality analysis must be done on the pull request before the changes can be merged.

☐ Restrict code coverage
Enforce minimum line coverage thresholds on pull requests. When configured, uploaded coverage data must meet the specified criteria before changes can be merged.

☐ Automatically request Copilot code review
Request Copilot code review for new pull requests automatically if the author has access to Copilot code review and their premium requests quota has not reached the limit.

Save changes
Revert changes

4. PED v1.0 and Baseline Gate/Sign-Off Evidence

Document Version History:

Version	Date	Author	Reviewers	Description
---------	------	--------	-----------	-------------

0.1	07/09/2026	All team members	—	Initial PED structure created
0.2	07/09/2026	Chanice Grant	Megan Wathen, Milenke Field	Engineering Foundation drafted, Problem & Business Need Identified
0.3	08/09/2026	Megan Wathen	Milenke Field, Chanice Grant	Risk Register and Forward Engineering Considerations added
0.4	08/09/2026	Milenke Field	Megan Wathen, Chanice Grant	FRs, NFRs, Acceptance Criteria, RTM added
1.0	Pending	All team members	All team members	Formal M1 baseline

Baseline Gate & Sign-Off Evidence

Functional & Quality Gate Milestones:

Gate Verification Item	Status (Pass/Fail)	Evidence/Artifact Link	Verified By	Date
Architecture Review	To Be Determined	Not yet created	To Be Determined	To Be Determined
Security & Pen Test	To Be Determined	Not yet created	To Be Determined	To Be Determined
Load & Scale Testing	To Be Determined	Not yet created	To Be Determined	To Be Determined

Gate Outcome is Accepted and ready to move to M2.

Formal Sign-Off Matrix:

FRs, NFRs, Acceptance Criteria, RTM Ownership Sign-Off

Approved by: Milenke Field

Digital Signature Identity: 578122@student.belgiumcampus.ac.za

Date: 08 Sep 2026

Engineering Foundation Drafted Sign-Off

Approved by: Chanice Grant

Digital Signature Identity: 601597@student.belgiumcampus.ac.za

Date: 08 Sep 2026

Risk Register and Forward Engineering Considerations Sign-Off

Approved by: Megan Heather Wathen

Digital Signature Identity: 601611@student.belgiumcampus.ac.za

Date: 08 Sep 2026

APPENDIX A – AI RESEARCH & VERIFICATION RECORD

<i>Team Member</i>	<i>AI/Tool Use</i>	<i>Purpose</i>	<i>Output Used?</i>	<i>How Independently Verified?</i>	<i>What was changed/rejected/accepted?</i>
Chanice Grant	ChatGPT	Used to assist with outlining and structuring the Engineering Foundation; identify relevant academic research; cross-reference the Master Project Brief and M1 requirements ; and support development of stakeholder, scope, constraint and decision-log analysis.	Partially	Outputs were checked against the CivicConnect Master Project Brief and Milestone 1 Brief. Suggested academic sources were located through Google Scholar and checked for relevance before use.	Relevant structural and research suggestions were accepted and adapted.
Megan Watheon	ChatGpt	Documentation and Planning: AI was used to assist with spelling and grammar, formulate work plans, and suggest table structures for organising independently researched project information.	Partially	All factual information entered into the tables was independently researched and reviewed by the team member. AI-generated wording and layouts were checked and modified where necessary before inclusion in the project documentation.	Table structures we accepted and adapted.

Milenk e Field	Google Search AI mode	To get a layout for the PED and Baseline Gate & Sign-Off Evidence section, To get table content for all aspects relating to FRs, NFRs, ACs, RTM, Traceability	Yes & Partially Yes	Compared with other academic/standards sources, recognised professional resources, and official technological documentation to check for alignment	Accepted the table format with header titles, Relevant accepted as is and Modified some of the content obtained as well but humanly verified
Milenk e Field	ChatGP T	To generate an initial assignment outline, assist with outlining and structuring, To get table content for all aspects relating to FRs, NFRs, ACs, RTM, Traceability	Yes & Partially Yes	Compared with other academic/standards sources, recognised professional resources, and official technological documentation to check for alignment	Table titles were accepted for formatting purposes, Relevant accepted as is and Modified some of the content obtained as well but humanly verified

References

Alzayed, A., 2024. Evaluating the role of requirements engineering practices in the sustainability of electronic government solutions. *Sustainability*, 16(1), p. 433.

Belgium Campus iTversity, 2026. *WEEK 1 — DAY 3 — SECTION 2 TRACEABILITY & CONTINUOUS VERIFICATION FOUNDATIONS*, Pretoria: Belgium Campus iTversity.

Castillo de Mesa, J., Méndez-Domínguez, P. & García, T. P., 2026. Organizational Barriers to Access in Community Social Services: Fragmentation, Bureaucratization, Congestion and the Role of Triage. *Social Sciences*, 15(7), p. 483.

Demir, M. O., Chouseinoglou, O. & Tarhan, A. K., 2024. Factors affecting architectural decision-making process and challenges in software projects: An industrial survey. *Journal of Software: Evolution and Process*, 36(10), p. e2703.

Gilal, R., Omar, M. & Rejab, M. M., 2023. The key factors contribute to time pressure in software development projects: A review. *Int J Adv Appl Sci*, 10(10), pp. 155-165.

Hartatik, 2022. APPLICATION OF DETERMINATION OF CULINARY BUSINESS OPPORTUNITIES STUDY WITH THE SAW METHOD AS A TOURIST ATTRACTION IN THE DIGITAL ERA. *International Journal of Economy, Education and Entrepreneurship*, 2(2), p. 488.

Ibrahim, A. K. & Ta'a, A. B., 2015. Mobile-Based Bus Ticketing System in Iraq. *European Journal of Computer Science and Information Technology*, 3(5), pp. 48-49.

Klasman, K., 2024. *How to prepare a system requirements table with the MoSCoW method.* [Online]

Available at: <https://www.novacura.com/how-to-define-system-requirements/>

[Accessed 7 Sept 2026].

Ray, F., 2026. *Non-Functional Requirements: Examples & Templates.* [Online]

Available at: <https://bettersoftware.uk/better-software-requirements/non-functional-requirements/>

[Accessed 7 Sept 2026].

Silva, A. S. et al., 2024. Mastering requirements volatility: Strategies for dynamic environments and multiple stakeholders in government software projects. *IEEE access*, Volume 12, pp. 183060-183077.